



# Latent Blackboards Are Easy to Ignore: A Diagnostic Study of Compressed Hidden-State Memory in a 2B Language Model Pair

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## Abstract

Continuous latent memories promise to let language models pass intermediate state without translating every step back into text. We study a concrete version of this idea: a small *latent blackboard* that compresses an upstream context into learned hidden vectors read by a downstream model. Across a controlled Phase-0 diagnostic suite on a same-architecture 2B mixed-attention checkpoint pair, we find that naive retrofit blackboards are constrained less by board slot count than by reader incentives. Exact token-level hidden-state reconstruction collapses under even 2:1 compression. Gist-style boards recover roughly 60–70% of the value of full context, but this recovery is nearly flat in the number of board slots and can be overstated by reader drift. We introduce two paired diagnostics, *board lift* and *LM drift*, and show that naive reader co-training mostly improves the no-board reader rather than board use. A no-board anchoring intervention changes the decomposition: board lift rises from 0.14–0.16 to 1.31–1.54 PPL points, while LM drift falls from about 1.18 to 0.20–0.32; the same direction holds in log-PPL space. The resulting picture is neither a failure of latent memory nor a solved system: retrofitted blackboards behave as lossy notes, and they are used only when the objective structurally discourages ignoring them. These results motivate from-scratch blackboard pretraining with forced cross-segment dependency.

**Keywords.** Latent memory; blackboard architectures; hidden-state compression; language model diagnostics; long-context modeling; reader drift.

## 1 Introduction

Modern Transformer language models communicate through tokens, yet their internal computation is continuous [7]. This mismatch motivates a simple question: can one model write a compact sequence of hidden vectors that another model reads as working memory? A successful answer would enable serial expert systems that pass latent state rather than verbose text, and it would provide a memory substrate distinct from a KV cache, which records the full token-level computation trace.

This paper studies the negative and positive parts of that question in CANCELI, a serial relay architecture. An upstream expert encodes a context; a learned resampler compresses the resulting hidden states into  $M$  board vectors; a downstream expert predicts the continuation from those board vectors. The desired board is not necessarily a lossless reconstruction of every token state. It may instead act as a continuous note: a compact state that preserves information useful for the next segment.

Our central finding is that optional latent boards are easy for pretrained readers to ignore. Many apparent gains disappear when we compare the same trained reader with and without the board. In other words, the reader can reduce the training loss through ordinary language-model adaptation rather than through board usage. Once we add a no-board anchor that constrains this escape route, the reader begins to use the board: board lift increases by roughly an order of magnitude, and the recovered value becomes mostly attributable to the board rather than no-board LM drift.

The paper is intentionally scoped as a diagnostic study, not as a claim that a final blackboard language model has been trained. Its contributions are:

- A controlled retrofit testbed for compressed latent blackboards: exact hidden-state reconstruction, embedding-level gist boards, layer- $K$  boards, last- $k$  truncation baselines, LoRA adaptation, reader co-training, and no-board anchoring.
- A negative result for lossless-style hidden-state compression: 2:1 compression already raises median student/teacher PPL ratio to 2.67, and stronger compression collapses by orders of magnitude.
- A diagnostic decomposition that separates actual board usage from no-board language-model drift.
- Evidence that the board can be read when the no-board shortcut is constrained, supporting from-scratch training with structural dependency as the next design step.

## 2 Setup and Diagnostics

### 2.1 Relay task

Let  $x$  be a context segment and  $y$  be a continuation segment. An upstream model  $A$  maps  $x$  to hidden states. A resampler  $C_\theta$  maps the hidden sequence to a board  $b \in \mathbb{R}^{M \times d}$ . A downstream model  $B$  predicts  $y$  from either the full context, the board, or no context. Most gist-style experiments use context length 448, continuation length 64, and a layer split  $K = 6$ . Unless stated otherwise, reported values are medians over 16 held-out evaluation windows.

The retrofit experiments use a same-architecture 2B checkpoint pair: a base-style upstream expert and an instruction-style downstream expert. The model has 24 layers, hidden size 2048, and a mixed attention stack with full-attention layers at indices 3, 7, 11, 15, 19, 23. LoRA is applied only to these full-attention projections in the LoRA variant; SSM-style layers are left untouched for stability.

### 2.2 Metrics

For a fixed continuation  $y$ , we evaluate

$$\text{PPL}_{\text{full}} = \text{PPL}(y \mid x), \quad \text{PPL}_{\text{gist}} = \text{PPL}(y \mid C_\theta(x)), \quad \text{PPL}_{\text{none}} = \text{PPL}(y \mid \emptyset). \quad (1)$$

The operational recovery statistic is

$$R = \frac{\text{PPL}_{\text{none}} - \text{PPL}_{\text{gist}}}{\text{PPL}_{\text{none}} - \text{PPL}_{\text{full}}}. \quad (2)$$

Here  $R = 1$  means the board closes the measured gap between no context and full context, while  $R = 0$  means it provides no benefit over no context. Because PPL is nonlinear, we use  $R$  as a practical diagnostic rather than as an information-theoretic quantity.

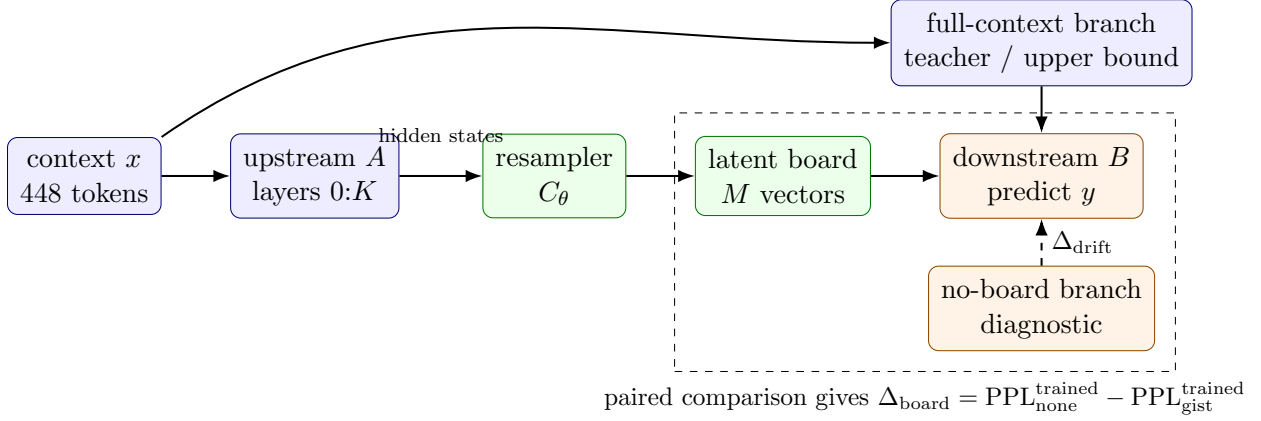


Figure 1: Evaluation design. A latent board is useful only if it improves the same trained reader relative to its no-board branch. This paired comparison separates board lift from ordinary no-board language-model drift.

For reader co-training, recovery can be misleading because the trained reader may improve without the board. We therefore decompose improvement relative to a frozen no-board baseline:

$$\Delta_{\text{board}} = \text{PPL}_{\text{none}}^{\text{trained}} - \text{PPL}_{\text{gist}}^{\text{trained}}, \quad (3)$$

$$\Delta_{\text{drift}} = \text{PPL}_{\text{none}}^{\text{frozen}} - \text{PPL}_{\text{none}}^{\text{trained}}. \quad (4)$$

Board lift is the paired gain from adding the board to the same trained reader. LM drift is the gain the reader obtains without the board. Because PPL differences are scale-sensitive, we also report a log-PPL sanity check for the co-training decomposition:

$$\Delta_{\text{board}}^{\log} = \log \text{PPL}_{\text{none}}^{\text{trained}} - \log \text{PPL}_{\text{gist}}^{\text{trained}}, \quad (5)$$

$$\Delta_{\text{drift}}^{\log} = \log \text{PPL}_{\text{none}}^{\text{frozen}} - \log \text{PPL}_{\text{none}}^{\text{trained}}. \quad (6)$$

We use this only as a robustness check on the paired decomposition, while retaining the original per-window median recovery statistic for the main gist and truncation curves.

### 2.3 Protocol suite

The Phase-0 suite is organized as a sequence of diagnostic probes rather than as one monolithic benchmark. Each protocol changes one part of the relay system—the compression target, the board frame, the reader adaptation path, or the no-board control—so that later results can be interpreted as failure-mode tests instead of isolated scores. Table 1 summarizes these probes and the local artifact used for each analysis.

Table 1: Phase-0 diagnostic protocols and their purpose. Each protocol isolates a different possible failure mode of latent blackboard retrofit.

| Protocol                  | Question                                                                         | Primary artifact                |
|---------------------------|----------------------------------------------------------------------------------|---------------------------------|
| Phase 0 reconstruction    | Can a short board losslessly reconstruct layer- $K$ hidden states?               | <code>phase0_curve.csv</code>   |
| Phase 0b gist             | Can an embedding-frame board preserve context value for continuation?            | <code>phase0b_gist.csv</code>   |
| Phase 0b + LoRA           | Does lightweight reader adaptation break the plateau?                            | <code>phase0b_lora.csv</code>   |
| Phase 0c truncation       | How much context value is in the last $k$ raw vectors, and how large is leakage? | <code>phase0c_trunc.csv</code>  |
| Phase 0d layer- $K$ codec | Does a trained codec beat last- $k$ truncation in the same frame?                | <code>phase0d_layerk.csv</code> |
| Phase 0e/f co-training    | Does an unfrozen reader use the board or drift without it?                       | <code>phase0f_grid.csv</code>   |

## 3 Results

### 3.1 Lossless-style hidden-state compression collapses

Figure 2 and Table 2 show that exact layer- $K$  hidden-state reconstruction is a poor target for aggressive board compression. The identity-like  $M = 512$  setting is healthy, with median student/teacher PPL ratio 1.07. But at  $M = 256$ , only 2:1 compression, the median ratio jumps to 2.67. At 4:1 and beyond, PPL collapses by orders of magnitude. This does not rule out blackboards, but it indicates that a compact board should not be evaluated as a lossless hidden-state video codec.

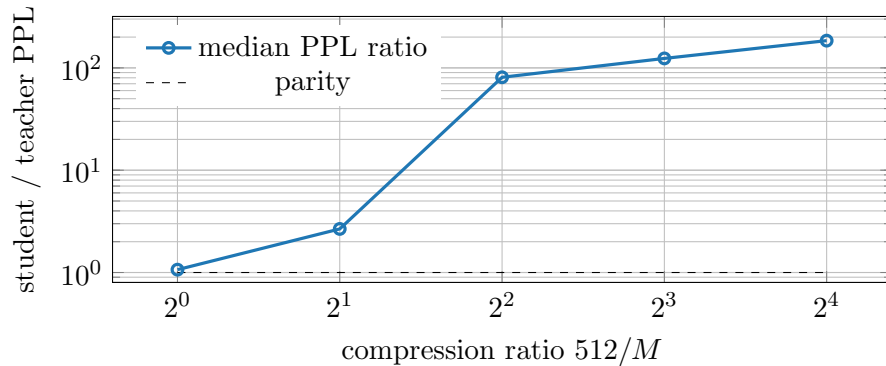


Figure 2: Exact hidden-state reconstruction becomes brittle under mild compression. A board that behaves like a lossy note is more realistic than a board that reconstructs every token state.

Table 2: Phase 0 hidden-state reconstruction. Teacher PPL is 2.83 across rows; all values are medians over 16 held-out windows.

| $M$ | Compression | Student PPL | Median ratio |
|-----|-------------|-------------|--------------|
| 512 | 1:1         | 3.02        | 1.07         |
| 256 | 2:1         | 8.70        | 2.67         |
| 128 | 4:1         | 253.44      | 80.95        |
| 64  | 8:1         | 441.57      | 123.74       |
| 32  | 16:1        | 837.69      | 184.66       |

### 3.2 Gist boards recover context value but plateau

Embedding-frame gist boards recover a substantial fraction of context value. With a frozen reader,  $M = 8$  gives  $R = 0.654$  at 56:1 compression, and  $M = 32$  gives  $R = 0.658$  at 14:1. Increasing  $M$  does not reliably improve recovery. LoRA adaptation improves the best  $M = 32$  result to  $R = 0.697$ , but  $M = 64$  remains at  $R = 0.688$ . The flatness suggests that the immediate bottleneck is not the slot count but the reader’s ability or incentive to use the soft board.

Figure 3 compares embedding gist boards with a clean last- $k$  truncation baseline. The truncation curve is steep: raw last- $k$  vectors improve as  $k$  grows and need roughly half of the original context vectors to approach full recovery. This implies that context information is distributed across many positions, not concentrated in a tiny suffix. However, the trained layer- $K$  codec dominates truncation at high compression when evaluated in the same frame (Table 3).

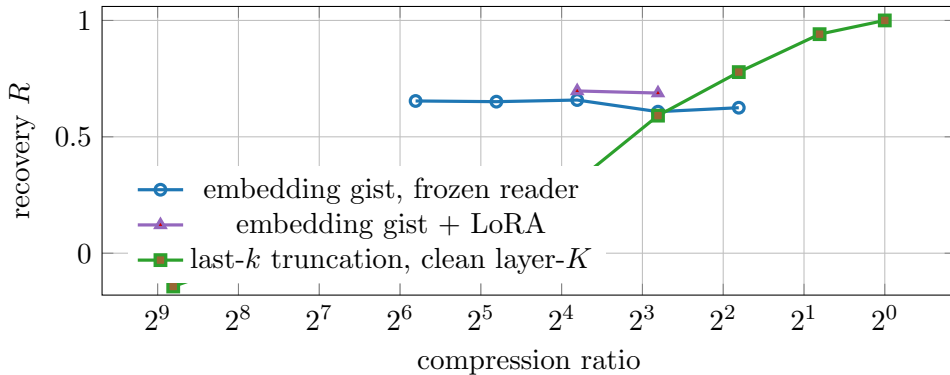


Figure 3: Gist boards are useful even at high compression but nearly flat in  $M$ . Raw last- $k$  truncation improves steeply as more original vectors are retained.

Table 3: Layer- $K$  codec vs. clean last- $k$  truncation under the same frame and scoring rule. The codec wins at high compression but plateaus near  $R \approx 0.55$ .

| $M$ or $k$ | Compression | Codec PPL | Codec $R$ | Trunc. $R$ |
|------------|-------------|-----------|-----------|------------|
| 8          | 56:1        | 5.26      | 0.475     | 0.078      |
| 16         | 28:1        | 5.08      | 0.579     | 0.189      |
| 32         | 14:1        | 5.08      | 0.544     | 0.310      |
| 64         | 7:1         | 5.31      | 0.563     | 0.591      |
| 128        | 3.5:1       | 5.31      | 0.540     | 0.778      |

### 3.3 Protocol leakage can overstate recovery

The truncation baseline exposes a protocol hazard. In the clean cross-checkpoint protocol,  $k = 32$  gives  $R = 0.310$ . In the contaminated protocol, where continuation states can already see the context before the board is inserted, the same  $k$  gives  $R = 0.757$ . At  $k = 64$ , clean recovery is 0.591 while contaminated recovery is 0.926. Thus, a large fraction of apparent success can come from letting continuation representations leak context before the board-reading stage. Decisive comparisons should therefore use clean continuation encoding.

### 3.4 Naive co-training learns the no-board shortcut

Reader co-training initially appears promising if measured only against frozen baselines. However, board lift reveals that the model is mostly improving without the board. With clean KL-to-teacher training at  $M = 32$ , the trained reader has  $\text{PPL}_{\text{gist}} = 5.526$  and  $\text{PPL}_{\text{none}} = 5.668$ , so the paired board lift is only 0.142. The same trained reader has LM drift 1.196 relative to the frozen no-board PPL 6.864. Most of the apparent improvement therefore comes from no-board language-model adaptation. In log-PPL space, the same comparison gives only 0.025 nats of board lift versus 0.192 nats of LM drift.

This diagnosis explains why simply unfreezing the reader does not break the plateau. The optimization problem contains an easy route: improve continuation modeling directly, and ignore a difficult latent side channel. A standard KL teacher on the board condition does not remove this route.

### 3.5 No-board anchoring forces actual board use

No-board anchoring changes the decomposition. Figure 4 shows the grid over anchor weights and board sizes. Without anchoring, board lift remains near 0.15 for  $M \in \{16, 32, 64\}$  while LM drift is around 1.13–1.20. With anchor weight 2, board lift rises to 1.31–1.41 and drift falls to 0.23–0.32. With anchor weight 5, board lift reaches 1.54 at  $M = 16$  and drift remains below 0.30 across the grid, though some larger-board settings slightly degrade, suggesting that too much anchoring can reduce adaptation capacity.

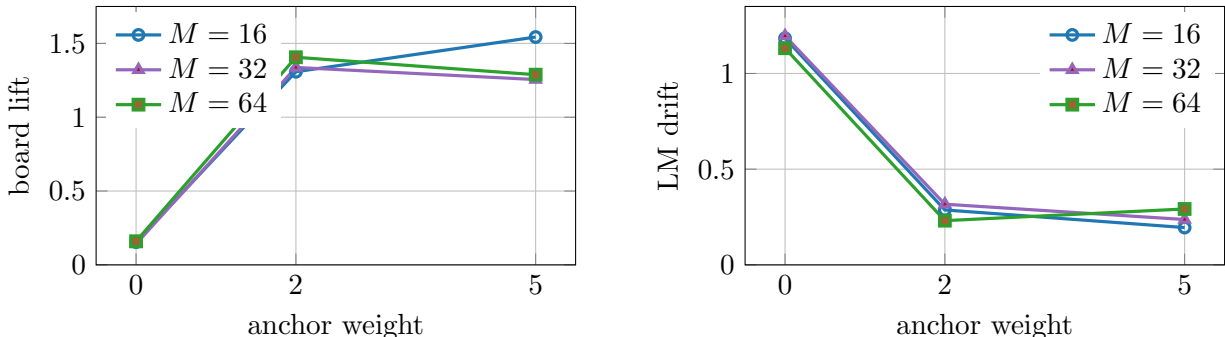


Figure 4: No-board anchoring converts apparent recovery into actual board usage. The dominant effect is anchor weight, not changing  $M$  from 16 to 64.

Figure 5 shows the same point as a decomposition for  $M = 32$ . With KL only, only about 11% of the PPL improvement is attributable to the board. With anchoring, most of the improvement is board lift. The log-PPL check gives the same conclusion: at anchor 2, board lift is 0.229 nats while LM drift is only 0.047 nats. The board is still not lossless: anchored recovery remains around

0.57–0.60, far from full-context performance. But the failure mode has changed from “the reader ignores the board” to “the board is a lossy gist memory.”

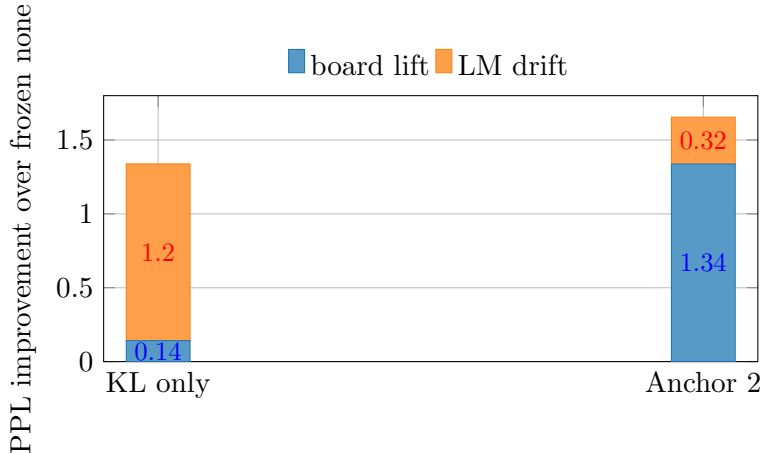


Figure 5: Apparent improvement decomposed for  $M = 32$ . KL-only co-training mostly improves the no-board reader; no-board anchoring makes the board responsible for most of the gain.

Table 4: Reader co-training decomposition. PPL-space and log-PPL-space diagnostics agree that anchoring shifts improvement from no-board drift to board usage.

| Condition   | $M$ | Anchor | $\text{PPL}_{\text{gist}}$ | $\Delta_{\text{board}}$ | $\Delta_{\text{drift}}$ | $\Delta_{\text{board}}^{\log} / \Delta_{\text{drift}}^{\log}$ |
|-------------|-----|--------|----------------------------|-------------------------|-------------------------|---------------------------------------------------------------|
| KL only     | 32  | 0      | 5.526                      | 0.142                   | 1.196                   | 0.025 / 0.192                                                 |
| Anchor grid | 32  | 2      | 5.209                      | 1.337                   | 0.317                   | 0.229 / 0.047                                                 |
| Anchor grid | 16  | 5      | 5.126                      | 1.543                   | 0.195                   | 0.263 / 0.030                                                 |

## 4 Related Work

**Long-context recurrence and compressed memory.** Transformer-XL extends attention beyond a fixed window by reusing hidden states from previous segments [2]. Compressive Transformers add a learned compression path for older activations [6]. These systems treat memory as an extension of the model’s own context stream. Our diagnostic setting is different: a board written by one checkpoint is read by another same-architecture checkpoint, and the question is whether a compact latent side channel is actually used by a pretrained reader.

**Memory tokens and recurrent segment models.** Recurrent Memory Transformers introduce learned memory tokens that carry state across segments [1]. The Stage-1 direction suggested by our results is close in spirit: cross-segment information should be forced through a memory interface. The present paper, however, focuses on why a memory-like interface fails when retrofitted onto an already competent reader with a no-board shortcut.

**Prompt compression and latent resampling.** Gist-token methods train compact prompt tokens that stand in for longer inputs [5]. Perceiver-style architectures use cross-attention to resample large inputs into compact latent arrays [4]. Our resampler plays a similar compression

role, but the main contribution is diagnostic: we separate board usage from no-board reader drift and show that optional soft boards are ignored unless the objective discourages that shortcut.

**Parameter-efficient adaptation.** LoRA adapts large models through low-rank updates [3]. In our mixed-attention checkpoint pair, LoRA on full-attention projections improves gist recovery only modestly and does not remove the plateau. This supports the broader conclusion that reader incentives and bypass paths matter more than small changes in board capacity or partial reader adaptation.

## 5 Discussion

**A blackboard is not a hidden-state video codec.** The exact reconstruction experiment fails under mild compression, and the truncation baseline shows that raw useful information is distributed over many context positions. This makes lossless relay a poor target for small  $M$ . A compact latent board is better understood as a learned note than as a faithful recording.

**The reader objective matters more than slot count.** Across frozen gist boards, LoRA, layer- $K$  codecs, and anchored co-training, performance is much flatter in  $M$  than a capacity-only story would predict. The large effect is whether the reader is forced to depend on the board. This is a training-objective and architecture issue, not just a resampler-size issue.

**Why retrofit is hard.** Retrofitting makes the board optional for a reader that already has strong continuation priors. The no-board branch can improve without learning to interpret the board, and recovery measured against a frozen baseline can hide this behavior. The no-board anchor works because it removes the most direct route for this drift.

**Implication for from-scratch training.** A from-scratch design can remove the pretrained no-board prior by making the board the only cross-segment path: segment  $t$  should not see segment  $t - 1$  tokens directly, only  $\text{board}_{t-1}$ . Stage-1 training should therefore combine physical bypass blocking, no-board anchoring or an equivalent constraint, and tasks where cross-segment information is actually needed.

## 6 Limitations and Submission Readiness

This is a Phase-0 diagnostic, not a finished blackboard language model. The study uses one 2B mixed-attention model family, one main split layer ( $K = 6$ ), a 100MB training subset, and medians over 16 evaluation windows. Results should be replicated with more seeds, more windows, additional split layers, and distinct model families. We add a log-PPL sanity check for the co-training decomposition, but a stronger version should compute token-level NLL differences directly from saved per-window losses. The strongest positive evidence is an anchoring intervention in a retrofit setting; it supports, but does not yet prove, that a from-scratch blackboard-pretrained model will scale. Finally, the experiments evaluate continuation perplexity rather than downstream reasoning, retrieval, or multi-expert collaboration tasks.

For a workshop or arXiv technical report, the present evidence supports a clear claim: *optional latent blackboards are ignored by pretrained readers unless the objective blocks no-board shortcuts*. For a stronger conference submission, the next experiments should add at least three seeds, a larger

held-out set, a second split layer such as  $K = 12$ , and a small from-scratch recurrent-board prototype demonstrating positive board lift without retrofitted anchoring.

## 7 Conclusion

Compressed latent blackboards can carry useful context, but pretrained readers tend to ignore them unless the training setup removes easier alternatives. Exact hidden-state compression fails quickly, and naive retrofit methods plateau. The decisive diagnostic is to separate board lift from LM drift: once the no-board escape route is constrained, board lift rises sharply and becomes the main source of improvement. The next step is therefore not a larger retrofit codec, but a from-scratch architecture and objective in which cross-segment information must flow through the board.

## References

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## A Local Artifacts Used in This Draft

The numerical claims in this draft are derived from local scripts, CSV outputs, and the Phase-0 handoff manual in the workspace. The most important artifacts are:

- `cancri_phase0b.gist.py`: embedding-frame gist-board training and the LoRA variant.
- `cancri_phase0c.trunc.py`: clean vs. contaminated last- $k$  truncation baselines.

- `cancri_phase0d_layerk_codec.py`: layer- $K$  codec protocol matching the truncation frame.
- `cancri_phase0e_unfreeze.py`: reader co-training, KL teacher, no-board anchor, and anchor grid.
- Reconstruction and gist CSVs: `phase0_curve.csv`, `phase0b_gist.csv`, `phase0b_lora.csv`, and `phase0c_trunc.csv`.
- Layer- $K$  and KL CSVs: `phase0d_layerk.csv` and `phase0e_clean_kl.csv`.
- Anchor CSVs: `phase0f_anchor.csv` and `phase0f_grid.csv`.
- Derived log-PPL summary: `phase0_nll_summary.csv`.

## B Command-Level Reproducibility

The following commands are written with relative paths only. Model and data directories should be placed under the project root or symlinked there before running the GPU experiments.

**Compile this draft.**

```
pdflatex -interaction=nonstopmode main.tex
bibtex main
pdflatex -interaction=nonstopmode main.tex
pdflatex -interaction=nonstopmode main.tex
```

**Smoke-test local scripts.**

```
python -X utf8 cancri_blackboard_distill.py --mode smoke
python -X utf8 cancri_phase0b_gist.py --mode smoke
python -X utf8 cancri_phase0c_trunc.py --mode smoke
python -X utf8 cancri_phase0e_unfreeze.py --mode smoke
```

**Representative Phase-0 reconstruction run.**

```
python -X utf8 cancri_blackboard_distill.py --mode train \
  --model-a Qwen3.5-2B-Base \
  --model-b Qwen3.5-2B-Instruct \
  --device cuda --dtype bfloat16 \
  --data phase0_subset.jsonl \
  --k 6 --seq-len 512 --batch-size 8 --steps 500 \
  --m-list 512,256,128,64,32 --keep-prefix 0 \
  --out-csv phase0_curve.csv
```

**Representative gist and truncation runs.**

```
python -X utf8 cancri_phase0b_gist.py --mode train \
  --model-a Qwen3.5-2B-Base \
  --model-b Qwen3.5-2B-Instruct \
  --device cuda --dtype bfloat16 \
```

```

--data phase0_subset.jsonl \
--ctx-len 448 --cont-len 64 --batch-size 8 --steps 600 \
--m-list 8,16,32,64,128 --out-csv phase0b_gist.csv

python -X utf8 cancri_phase0c_trunc.py --mode run \
--model-a Qwen3.5-2B-Base \
--model-b Qwen3.5-2B-Instruct \
--device cuda --dtype bfloat16 \
--data phase0_subset.jsonl --k 6 --out-csv phase0c_trunc.csv

```

### Representative no-board anchor grid.

```

python -X utf8 cancri_phase0e_unfreeze.py --mode train \
--model-a Qwen3.5-2B-Base \
--model-b Qwen3.5-2B-Instruct \
--device cuda --dtype bfloat16 \
--data phase0_subset.jsonl \
--ctx-len 448 --cont-len 64 --k 6 --batch-size 8 \
--steps 1000 --b-lr 2e-5 --kl-teacher --kl-weight 1.0 \
--beta 0.5 --anchor-list 0,2,5 --m-list 16,32,64 \
--out-csv phase0f_grid.csv

```

## C Reproducibility Checklist

Table 5: Submission-oriented reproducibility checklist.

| Item            | Status in this draft                                                                                                              |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Model family    | Same-architecture 2B mixed-attention Base/Instruct pair; layer split $K = 6$ .                                                    |
| Data            | 100MB Phase-0 subset with 448-token context and 64-token continuation for gist-style runs.                                        |
| Metrics         | PPL-space recovery, paired board lift, LM drift, and log-PPL sanity checks for co-training.                                       |
| Evaluation size | Medians over 16 held-out windows; this is sufficient for diagnosis but not final statistical confidence.                          |
| Seeds           | Not yet replicated across multiple random seeds.                                                                                  |
| Baselines       | Full context, no context, clean vs. contaminated truncation, frozen reader, LoRA reader, unfrozen reader, and anchored reader.    |
| Compute         | GPU runs were designed for a ROCm/MI300X-class environment; CPU smoke tests validate plumbing only.                               |
| Known gaps      | More seeds, more evaluation windows, additional split layers, token-level NLL logs, and a from-scratch recurrent-board prototype. |